

## 1. Executive Summary

|                         |         | Traffic Summary        |  | Count      |  |
|-------------------------|---------|------------------------|--|------------|--|
| Metric                  | Value   | Total Flows Observed   |  | 83408      |  |
| Total Threats Mitigated | 14      | Total Packets Analyzed |  | 3570395086 |  |
| High Priority           | 7       | Malicious Packets      |  | 3557133737 |  |
| Low Priority            | 7       | True Negatives Passed  |  | 71399      |  |
| Manual Releases         | 0       |                        |  |            |  |
| Manual Blocks           | 0       | Attack Vector          |  | Count      |  |
|                         |         | ICMP Flood             |  | 11         |  |
| ML Metrics              | Value   | SYN Flood              |  | 0          |  |
| False Positives         | 0       | UDP Flood              |  | 0          |  |
| FP Rate                 | 0.00%   | Uncertain              |  | 3          |  |
| IF Precision            | 100.00% |                        |  |            |  |
| IF Recall               | 97.46%  | Action                 |  | Count      |  |
| IF F1-Score             | 98.71%  | Quarantined            |  | 3          |  |
| IF Accuracy             | 99.63%  | Time Ban               |  | 3          |  |
|                         |         | Blackhole              |  | 0          |  |
|                         |         | Released               |  | 2          |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 97.46%  | Share of actual attacks successfully flagged         |
| F1-Score                  | 98.71%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.63%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 2.54%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 97.46%  | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 11982          | FN 312            |
| Actual: Normal | FP 0              | TN 71087          |

### 2b. Random Forest: Attack Classification

| Metric    | Value   | Description                                                  |
|-----------|---------|--------------------------------------------------------------|
| Precision | 100.00% | Share of classified flows assigned the correct attack type   |
| Recall    | 100.00% | Share of attacks of each type that were correctly identified |
| F1-Score  | 100.00% | Balanced measure combining Precision and Recall              |
| Accuracy  | 100.00% | Overall proportion of correct classifications                |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 0              | 0               | 0              |
| Act: ICMP | 0              | 11982           | 0              |
| Act: UDP  | 0              | 0               | 0              |

### 2c. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 469.9 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 13.7 ms  | Interval between the anomaly flag and the FlowMod blocking rule install |

2d. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 1.87% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 2.11% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 3.46% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.24 | 3        | 3            | 2026-09-16 12:58:49 | 2026-09-16 13:00:11 |
| 10.0.0.23 | 2        | 2            | 2026-09-16 12:58:47 | 2026-09-16 12:59:58 |
| 10.0.0.25 | 2        | 2            | 2026-09-16 12:58:47 | 2026-09-16 12:59:58 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action        |
|---------------------|-----------|-------|------------|------------|----------|---------------|
| 2026-09-16 12:47:46 | 10.0.0.23 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-16 12:47:46 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-16 12:47:48 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-16 12:47:56 | 10.0.0.23 | DDoS  | ICMP Flood | 90.8%      | High     | Time Ban (1m) |
| 2026-09-16 12:47:56 | 10.0.0.25 | DDoS  | ICMP Flood | 91.2%      | High     | Time Ban (1m) |
| 2026-09-16 12:47:56 | 10.0.0.25 | DDoS  | ICMP Flood | 91.2%      | Critical | Time Ban      |
| 2026-09-16 12:47:57 | 10.0.0.23 | DDoS  | ICMP Flood | 90.8%      | High     | Time Ban      |
| 2026-09-16 12:47:58 | 10.0.0.24 | DDoS  | ICMP Flood | 91.1%      | High     | Time Ban (1m) |
| 2026-09-16 12:47:59 | 10.0.0.24 | DDoS  | ICMP Flood | 91.1%      | Critical | Time Ban      |
| 2026-09-16 12:48:57 | 10.0.0.23 | DDoS  | ICMP Flood | 93.8%      | High     | Released      |
| 2026-09-16 12:48:57 | 10.0.0.25 | DDoS  | ICMP Flood | 93.5%      | High     | Time Ban (2m) |
| 2026-09-16 12:48:59 | 10.0.0.24 | DDoS  | ICMP Flood | 94.2%      | High     | Released      |
| 2026-09-16 12:49:08 | 10.0.0.23 | DDoS  | ICMP Flood | 93.8%      | Critical | Time Ban (2m) |
| 2026-09-16 12:49:09 | 10.0.0.24 | DDoS  | ICMP Flood | 94.2%      | Critical | Time Ban (2m) |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.24 | ICMP Flood | 0.6460   | 94.2% | Critical | Phase 2 | 3        | 10s      | 2026-09-16 13:00:11 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6284   | 94.2% | High     | Phase 2 | 1        | 1m 11s   | 2026-09-16 13:00:00 | Ban Expired                               |
| 10.0.0.25 | ICMP Flood | 0.6418   | 92.1% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-16 12:59:58 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6413   | 92.1% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-16 12:59:58 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6453   | 91.1% | High     | Phase 2 | 1        | 11s      | 2026-09-16 12:59:00 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6454   | 91.1% | High     | Phase 2 | 1        | 11s      | 2026-09-16 12:58:58 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6454   | 90.9% | High     | Phase 2 | 1        | 11s      | 2026-09-16 12:58:58 | Attack persisted - escalating to time ban |

#### Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

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SIGNATURE OVER PRINTED NAME

**Network Administrator**

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DATE